Granulon: Awakening Pixel-Level Visual Encoders with Adaptive Multi-Granularity Semantics

To Reviewer #qAUj (score: 4):

Q1: *Differentiability of AdaTA clustering and gradient backpropagation.* **A1:** AdaTA is non-differentiable and excluded from optimization. We will clarify this wording.

Q2: *Dependency of controller training on GPT-4o annotations.* **A2:** ❶ The controller solves a **lightweight classification problem** which is easy. ❷ The downstream LLM in MLLM and even small model can do this. ❸ Qwen2.5-1.5B can reach $\sim 95\%$ accuracy. GPT-4o was used for convenience by API. We will clarify this in the appendix.

Q3: *Additional cost of AdaTA.* **A3:** Thanks for the comment. AdaTA overhead is minor compared to visual encoding and decoding: latency and FLOPs $< 5\%$, while yielding 30% performance gain. We will add this quantification.

To Reviewer #jn4j (score: 4):

Q1: *Limited novelty vs. prior query-adaptive or token compression methods.* **A1:** Unlike token compression field focusing on **semantic** token reduction & efficiency, Granulon bridges **pixel- and semantic-level** representations (Fig.1).

Figure 1. Comparison of token compression and `Granulon`.

❷ We novelty reveals **pixel-level** information as an effective and **underexplored new direction** for MLLMs field.

Q2: *Generality beyond DINO-family visual encoders.* **A2:** ❶ DINOv3 is among the first visual encoder showing strong emergent pixel-level capabilities (Fig.2), making it a suitable starting point. ❷ Our framework is **encoder-agnostic** and can **naturally extend** to future pixel-level encoders, which we view as an *exciting and open novelty direction.*

Figure 2. Attention maps of different visual encoders.

Q3: *Cost–benefit trade-off of adaptive controller and added complexity.* **A3:** ❶ We have conducted controlled comparisons across visual encoders with comparable tokens. ❷ Fig.7 shows additional overhead is minimal.

❸ In addition, we think Pixel-level granularity control also supports your concern about **token compression field** with comparable performance, indicating a promising direction.

To Reviewer #YQXk (score: 3):

Q1: *Ambiguity in diagram symbols and flow.* **A1:** We will revise all symbols and refine the algorithm for clearing.

Q2 & Q3 & Q4: *Training with loss update & K choice & LLM first layer as Controller.* **A2:** We first **clarify** that ❶ controller and clustering are fixed (non-trainable), with a **single forward pass**. ❷ K is chosen by accuracy–complexity trade-off (Fig.7). ❸ The controller solves a **lightweight classification problem** which is easy.

Q5: *Hyperparameters in Eq.(10).* **A3:** Eq.(10) aggregates spatial support, semantic coherence, and separability using **data-driven normalized weights**, rather than tunable hyperparameters. Formulations will be added to manuscript.

Q6: *Granularity mapping for ambiguous queries.* **A4:** Thanks for your careful observation. ❶ We explore a novel direction of enhancing semantic understanding in pixel-level visual encoders for MLLMs, rather than CLIP-style semantic encoders. ❷ More adaptive text–image conditioning can be built upon this foundation.

Q7.1: *Limits of text-driven pre-filtering.* **A5:** ❶ Controller is **model-agnostic**. ❷ Downstream LLM in MLLMs can also handle text input. ❸ Controller can be replaced by plug-and-play modules flexibly.

Q7.2: *Generality to larger backbones.* **A6:** *Following your comments*, we have added and observed consistent improvements in Table 1, supporting the expected trend.

Table 1. Comparison of visual encoders across reasoning tasks.

Encoders	Qwen2.5-7B		Qwen2.5-14B	
	Accuracy(%)↑	Hallucination(%)↓	Accuracy(%)↑	Hallucination(%)↓
`CLIP`	$35.82_{\uparrow 16.41}$	$62.12_{\downarrow 20.69}$	$37.29_{\uparrow 20.56}$	$60.77_{\downarrow 18.98}$
`DINOv3`	$40.07_{\uparrow 12.16}$	$48.62_{\downarrow 7.19}$	$51.14_{\uparrow 6.71}$	$46.98_{\downarrow 5.19}$
Ours	**52.23**	**41.43**	**57.85**	**41.79**

Q8 & Q9: *Architecture variants and native MLLMs.* **A7:** ❶ We target a novel pixel-to-semantic bridging direction and validate it on mainstream MLLMs. ❷ Therefore, we focuses on **diversity classic** architectures to ensure fair comparisons rather than variations. ❸ But Granulon is also directly integrated with variations without changes.

Table 2. Comparison of visual encoders with different sizes.

Encoders	A-OKVQA		Reasoning	
	BERTscore(%)↑	GPTscore(%)↑	BERTscore(%)↑	GPTscore(%)↑
`CLIP-L/14@336`	21.79	10.3	87.79	29.36
`Ours-s16`	54.22	33.9	88.76	48.51
`Ours-l16`	57.13	35.9	88.94	49.31
`Ours-7b16`	64.06	43.1	89.33	55.83

Q10: *Sensitivity to visual encoder variants.* **A8:** *Following your comments*, we have added Table 2, showing that `Granulon` can achieve higher scores in larger size.

Q11: *Ablations on granularity variants; criteria for kernel size and cluster number.* **A9:** Thanks for the question. Actually, Fig.6–7 have provided granularity ablations, where hyperparameters are chosen by accuracy–complexity trade-of. While optimal settings may vary, the qualitative trends are consistent across backbones.

Responses to all minor questions:

We add detailed `CLIP` configurations, metric definitions and evaluation prompts in the appendix.

We simplify Fig.1 and Fig.2 and clarify the terminology of granularity and implementation details in the paper.